POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

1091

Introduction to Polymer Science

Polymer Technology | Semester I | Practicum | Credits 4.5 | Contact Hours 90

A complete digital handbook covering history, classification, natural & synthetic rubbers, plastics and fibres — designed for first-semester diploma students.

Course Information

Course Description

Polymers are crucial to modern life, providing lightweight, durable, and versatile materials for everything from clothing and packaging to advanced medical devices, vehicles, and electronics, enhancing comfort, efficiency, and safety while enabling technological progress, though managing their waste through recycling and biodegradable alternatives is a growing focus. This course intends to give students an introduction to different types of polymers, their chemical structure, properties and applications. They also get an opportunity to prepare an album of different types of polymer products available in the market and find out the polymeric material with which the product is made.

Course Objectives

1. To summarise the history, general characteristic and classification of polymers.
2. To understand Natural rubber and different forms of natural rubber.
3. To understand about the commercially available Synthetic rubbers.
4. To understand about different types of plastics and fibres.

Course Pre-requisite

Topic	Course Code	Course Title	Semester
Hydrocarbons, alkane, alkene, alkynes	—	—	Secondary Education
Polymers	—	—	Secondary school

Course Outcomes (COs)

CO	Course Outcome	Bloom's Taxonomy Level
CO1	Explain the history, general characteristic and classification of polymers	Understand
CO2	Explain the structure, property and applications of Natural rubber	Understand
CO3	Comprehend different types of synthetic rubber, structural formula, properties and applications	Understand
CO4	Explain the classification, Structural formula, property and applications of Plastics and fibres	Understand

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	–	–	–	–	–	–	–	–	–	3
CO2	3	–	–	–	–	–	–	–	–	–	3
CO3	3	–	–	–	–	–	–	–	–	–	3
CO4	3	–	–	–	–	–	–	–	–	–	3
Total	12	–	–	–	–	–	–	–	–	–	12
Correlation Level	3	–	–	–	–	–	–	–	–	–	3

Legends: 1 – Low; 2 – Medium; 3 – High; No Correlation – "–"

Teaching and Assessment Scheme

Teaching Scheme / Week				Self-learning (SS) / Week	Total Hours / Semester	Credits C	Examination Scheme				Total
L	T	P	S				Theory CIE	Theory ESE	Practical CIE	Practical ESE	
3	0	3	0	6	90	4.5	20	60	20	40	140

L: Lecture (1 unit = 1 hour, 1 credit) · **T**: Tutorial · **P**: Practical (1 unit = 1 hour, 0.5 credit) · **S**: Project · **SS**: Self-Learning · **CIE**: Continuous Internal Evaluation · **ESE**: End Semester Examination

Module I — History of Polymers and Classification of Polymers

Instructional Hours: Theory 12 h · Practical 12 h · Mapped CO: CO1

1.1 History of Polymers

Polymers have been part of human life for thousands of years. Natural polymers such as cotton, wool, silk, wood and natural rubber were used long before their chemical nature was understood.

Key historical milestones

- **Ancient times**: Natural rubber (from Hevea brasiliensis) used by Mesoamerican civilisations for balls and waterproofing. Cotton, wool and silk used for textiles.
- **1839**: Charles Goodyear discovered vulcanisation of natural rubber with sulphur, making it useful over a wide temperature range.
- **1860s–1900s**: Cellulose nitrate (Parkesine, Celluloid) and regenerated cellulose (Rayon) marked the beginning of man-made polymers.
- **1907**: Leo Baekeland invented Bakelite – the first fully synthetic thermosetting plastic.
- **1920s–1930s**: Hermann Staudinger proposed the macromolecular hypothesis (long-chain molecules). Wallace Carothers developed Nylon (polyamide) at DuPont.
- **1930s–1950s**: Commercialisation of polyethylene, PVC, polystyrene, synthetic rubbers (SBR, NBR, CR) and acrylic fibres.
- **1950s onwards**: Stereoregular polymers (Ziegler–Natta catalysts), engineering plastics, high-performance fibres (carbon, aramid) and biomedical polymers.

Today polymers are indispensable in packaging, transportation, electronics, medicine, construction and consumer goods.

മലയാളം വിശദീകരണം

പോളിമറുകൾ ആയിരക്കണക്കിന് വർഷങ്ങളായി മനുഷ്യജീവിതത്തിന്റെ ഭാഗമാണ്. പ്രകൃതിദത്ത റബ്ബർ, പരുത്തി, കമ്പിളി, പട്ട് എന്നിവ പുരാതന കാലം മുതൽ ഉപയോഗിച്ചിരുന്നു. 1839-ൽ ചാൾസ് ഗുഡ്‌യിയർ

വൾക്കനൈസേഷൻ കണ്ടെത്തി. 1907-ൽ ബേക്കലൈറ്റ് എന്ന ആദ്യത്തെ പൂർണ്ണമായും കൃത്രിമ പ്ലാസ്റ്റിക് ഉണ്ടായി. ഇന്ന് പോളിമറുകൾ പാക്കേജിംഗ്, വാഹനങ്ങൾ, ഇലക്ട്രോണിക്സ്, മെഡിസിൻ എന്നിവയിൽ അത്യാവശ്യമാണ്.

1.2 Hydrocarbons

Definition: Compounds containing only carbon and hydrogen atoms.

Saturated Hydrocarbons

Contain only single bonds between carbon atoms (C–C). General formula for alkanes: C_nH_{2n+2}. They are less reactive.

Examples: Methane (CH_4), Ethane (C_2H_6), Propane (C_3H_8).

Unsaturated Hydrocarbons

Contain at least one double bond (alkenes) or triple bond (alkynes). More reactive than saturated hydrocarbons.

Examples: Ethene (C_2H_4), Propene (C_3H_6), Ethyne (C_2H_2).

Aliphatic Hydrocarbons

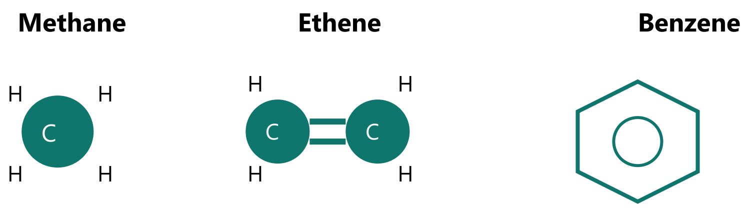
Open-chain (straight or branched) or non-aromatic cyclic compounds. Can be saturated or unsaturated.

Examples: Methane, ethane, cyclohexane, ethene.

Aromatic Hydrocarbons

Contain one or more benzene rings (planar, conjugated, obey Hückel's rule). Special stability due to delocalised π-electrons.

Examples: Benzene (C_6H_6), Toluene, Naphthalene.



Simple structural representations of methane, ethene and benzene

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ഹൈഡ്രോകാർബണുകൾ കാർബണും ഹൈഡ്രജനും മാത്രം അടങ്ങിയ സംയുക്തങ്ങളാണ്. സാച്ചുറേറ്റഡ് ഹൈഡ്രോകാർബണുകളിൽ ഒറ്റ ബോണ്ടുകൾ മാത്രം. അൺസാച്ചുറേറ്റഡ് ഹൈഡ്രോകാർബണുകളിൽ ഡബിൾ അല്ലെങ്കിൽ ട്രിപ്പിൾ ബോണ്ടുകൾ ഉണ്ട്. അലിഫാറ്റിക് തുറന്ന ശൃംഖലയും, അരോമാറ്റിക് ബെൻസീൻ വലയങ്ങളും ആണ്.

1.3 Homologous Series, Alkanes, Alkenes and Alkynes

Homologous series: A series of organic compounds having the same functional group and similar chemical properties, successive members differing by $-CH_2-$ (methylene) group.

Alkanes (Paraffins)

Saturated open-chain hydrocarbons. General formula C_nH_{2n+2}. Bond angle ≈ 109.5° (sp^3 hybridisation). Relatively inert.

Examples: CH_4, C_2H_6, C_3H_8, C_4H_{10}.

Alkenes (Olefins)

Contain one C=C double bond. General formula C_nH_{2n}. Planar around double bond (sp^2). Undergo addition reactions.

Examples: Ethene C_2H_4, Propene C_3H_6.

Alkynes

Contain one C≡C triple bond. General formula C_nH_{2n-2}. Linear geometry around triple bond (sp). Highly reactive.

Examples: Ethyne (acetylene) C_2H_2, Propyne C_3H_4.

> **Important**: Many monomers used in polymerisation are unsaturated (alkenes or dienes) because the double bond can open and form long chains.

1.4 Micromolecules and Macromolecules

Micromolecules: Small molecules with low molecular weight (usually < 1000 g/mol). Example: NaCl (58.5 g/mol), water, glucose, monomer units.

Macromolecules: Very large molecules with high molecular weight (thousands to millions g/mol) formed by covalent linking of many small units. Example: Natural rubber (polyisoprene, MW ≈ 10^5–10^6), proteins, cellulose, synthetic plastics.

The difference in molecular weight is the main reason for the vast difference in physical properties (strength, elasticity, melting behaviour).

1.5 Monomers, Oligomers, Polymers and Repeating Units

- **Monomer**: Small molecule that can join with other identical or different molecules to form a polymer. Examples: Ethene, vinyl chloride, isoprene, styrene.
- **Oligomer**: Short chain of a few monomer units (usually 2–10). Intermediate between monomer and polymer.
- **Polymer**: Large molecule made by covalent linking of many monomer units (degree of polymerisation usually > 100).
- **Repeating unit**: The structural unit that repeats along the polymer chain. For polyethylene the repeating unit is $–CH_2–CH_2–$.

$$n\ CH_2{=}CH_2 \rightarrow [-CH_2-CH_2-]_n$$

Monomer → Polymer (polyethylene)

1.6 Polymerisation and Degree of Polymerisation

Polymerisation: Chemical process in which monomer molecules join together to form a polymer chain (or network).

Two broad types:

- **Addition (chain-growth)**: Monomers add without loss of atoms (e.g., polyethylene, polystyrene, PVC).
- **Condensation (step-growth)**: Monomers join with elimination of small molecules such as water (e.g., nylon, polyester).

Degree of Polymerisation (DP): Average number of repeating units in a polymer chain.

$$\text{Molecular weight of polymer } (M_w) = DP \times \text{Molecular weight of repeating unit } (M_0)$$

$$\text{or} \quad DP = M_w / M_0$$

Worked Example 1

A polyethylene sample has average molecular weight 84 000 g/mol. Molecular weight of repeating unit $–CH_2–CH_2–$ is 28 g/mol. Find DP.

Solution: DP = 84 000 / 28 = 3000.

Worked Example 2

DP of a polystyrene sample is 500. Molecular weight of styrene repeating unit is 104 g/mol. Calculate average molecular weight.

Solution: M_w = 500 × 104 = 52 000 g/mol.

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പോളിമറൈസേഷൻ എന്നത് മോണോമർ തന്മാത്രകൾ ചേർന്ന് നീണ്ട ശൃംഖല ഉണ്ടാക്കുന്ന പ്രക്രിയയാണ്. ഡിഗ്രി ഓഫ് പോളിമറൈസേഷൻ (DP) എന്നത് ശൃംഖലയിലെ ആവർത്തന യൂണിറ്റുകളുടെ ശരാശരി എണ്ണമാണ്. M_w = DP

× M_0.

1.7 Classification of Polymers

Based on Origin

- **Natural polymers**: Occur in nature. Examples: Natural rubber, cellulose, starch, proteins, DNA, silk, wool.
- **Synthetic polymers**: Man-made by polymerisation. Examples: Polyethylene, PVC, Nylon, SBR, Bakelite.
- **Derived (semi-synthetic) polymers**: Chemically modified natural polymers. Examples: Cellulose acetate, vulcanised rubber, rayon.

Based on Backbone (Organic / Inorganic)

- **Organic polymers**: Main chain contains carbon atoms. Majority of commercial polymers (PE, PP, PVC, Nylon, polyester).
- **Inorganic polymers**: Main chain does not contain carbon (or carbon is not the primary atom). Example: Silicone rubber (–Si–O– backbone), polyphosphazenes.

Based on Chain Composition

- **Homochain polymers**: All atoms in the main chain are the same (usually carbon). Examples: Polyethylene, polystyrene.
- **Heterochain polymers**: Main chain contains more than one type of atom. Examples: Nylon (C, N), Polyester (C, O), Silicone (Si, O).

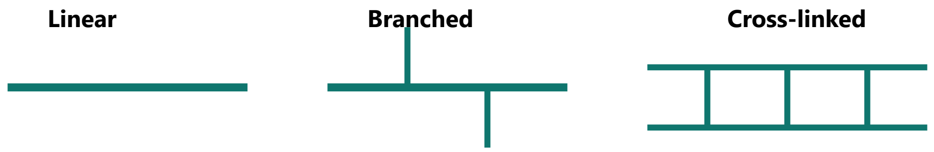
Based on Monomer Type

- **Homopolymer**: Made from only one type of monomer. Example: Polyethylene, PVC, natural rubber.
- **Copolymer**: Made from two or more different monomers.
  - **Random copolymer**: Monomers arranged randomly. Example: SBR (styrene + butadiene random).
  - **Alternating copolymer**: Monomers alternate regularly (A–B–A–B...).

Based on Structure (Architecture)

- **Linear polymers**: Long continuous chain without branches. Can pack well → higher crystallinity and density. Example: HDPE.

- **Branched polymers**: Side chains attached to main chain. Lower packing efficiency. Example: LDPE.
- **Cross-linked polymers**: Chains connected by covalent bridges forming a three-dimensional network. Do not melt or dissolve. Example: Vulcanised rubber, Bakelite, epoxy resins.



Schematic of linear, branched and cross-linked polymer chains

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പോളിമറുകളെ ഉത്ഭവം അനുസരിച്ച് പ്രകൃതിദത്തം, കൃത്രിമം, ഡെറൈവ്ഡ് എന്നിങ്ങനെ തരംതിരിക്കാം. ഘടന അനുസരിച്ച് ലീനിയർ, ബ്രാഞ്ച്ഡ്, ക്രോസ്-ലിങ്ക്ഡ്. ഹോമോപോളിമർ ഒരു തരം മോണോമർ മാത്രം; കോപോളിമർ രണ്ടോ അതിലധികമോ മോണോമറുകൾ.

Practical Module I — Specimen Preparation & Album

Objective: Collect samples of natural, synthetic, organic and inorganic polymers; record name, three properties and applications; prepare an album.

Suggested Samples

Category	Examples to Collect	Typical Properties	Applications
Natural	Cotton thread, wood piece, natural rubber band, silk thread	Biodegradable, renewable, variable strength	Textiles, paper, tyres, ropes
Synthetic	PE bag, PVC pipe piece, Nylon fishing line, Polyester fabric	Durable, processable, resistant to many chemicals	Packaging, pipes, clothing, engineering parts
Organic	PE, PP, Nylon, Polyester, PVC (as above)	Carbon-backbone, wide property range	Everyday plastics & fibres
Inorganic	Silicone rubber (kitchen spatula, sealant, medical tubing)	Heat resistant, flexible, biocompatible	Seals, medical devices, cookware

Observation Format (Album Page)

1. Photograph / glued sample
2. Name of polymer
3. Category (natural / synthetic / organic / inorganic)
4. Three key properties
5. Two important applications

6. Source of sample

Precautions

- Collect only clean, dry, non-hazardous samples.
- Label clearly; avoid confusion between similar-looking plastics.
- Do not collect sharp or chemically contaminated pieces without protection.

Module II — Natural Rubber

Instructional Hours: Theory 11 h · Practical 11 h · Mapped CO: CO2

2.1 Origin, Monomer and Structure of Natural Rubber

Origin: Natural rubber (NR) is obtained mainly from the latex of the rubber tree *Hevea brasiliensis*. Kerala is the leading rubber-producing state in India. The Rubber Board (headquartered at Kottayam) promotes research, development and marketing.

Monomer: Isoprene (2-methyl-1,3-butadiene), C_5H_8.

Polymer: cis-1,4-Polyisoprene. Almost 100 % of the double bonds in natural rubber have the cis configuration, giving high elasticity.

$$n\ CH_2{=}C(CH_3){-}CH{=}CH_2 \rightarrow [{-}CH_2{-}C(CH_3){=}CH{-}CH_2{-}]_n$$

(cis-1,4-polyisoprene)

Molecular weight: Typically 10^5 to several × 10^6 g/mol. High MW contributes to high tensile strength and elasticity.

cis-1,4-Polyisoprene repeating unit:

—CH

$_2$—C(CH_3)=CH—CH_2— (cis configuration)

2.2 Properties of Natural Rubber

1. High elasticity and resilience
2. Good tensile strength (especially after vulcanisation)
3. Excellent abrasion resistance
4. Low hysteresis (good for dynamic applications)
5. Permeable to gases (not suitable for pure air-retention without compounding)
6. Attacked by oils, solvents and ozone (needs protection)
7. Biodegradable under certain conditions

2.3 Latex — Field Latex, Concentrated Latex, Double Centrifuged Latex

Field latex: Fresh latex collected from the tree. Dry rubber content (DRC) ≈ 30–35 %. Contains proteins, resins, sugars and minerals. Unstable; coagulates on standing.

Concentrated latex: Field latex is concentrated by centrifuging or creaming to DRC ≈ 60 %. Easier to transport and process.

Double centrifuged latex: Centrifuged twice to remove more non-rubber constituents; higher purity, used for medical gloves and high-quality products.

Preservation of Latex

Ammonia is the most common preservative (0.2–0.7 %). It prevents bacterial action and coagulation. Other systems: low-ammonia with secondary preservatives (TMTD, zinc oxide, etc.).

Concentration Methods

- **Centrifuging**: High-speed centrifuge separates rubber particles (lighter) from serum.
- **Creaming**: Addition of creaming agents (e.g., ammonium alginate) causes rubber particles to rise.

2.4 Forms of Dry Rubber — RSS, TSR, ISNR

- **RSS (Ribbed Smoked Sheet)**: Coagulated latex is sheeted, ribbed and smoked. Graded visually (RSS 1 to RSS 5). Traditional form.
- **TSR (Technically Specified Rubber)**: Graded by technical specifications (dirt, ash, nitrogen, volatile matter, plasticity). More consistent quality. Includes ISNR grades.
- **ISNR (Indian Standard Natural Rubber)**: Indian grades of TSR produced under Bureau of Indian Standards specifications. Common grades: ISNR 5, 10, 20, 50.
- Other forms: Crepe rubber (pale crepe, brown crepe), ADS (Air Dried Sheet).

Outline of RSS Production

1. Dilution and sieving of latex
2. Coagulation with formic or acetic acid
3. Milling into sheets on sheeting batteries
4. Ribbing
5. Smoking in smoke house (4–7 days)
6. Grading and packing

ISNR / TSR Production (simplified)

1. Coagulation → crumb formation (or continuous process)
2. Washing and drying
3. Blending to meet technical specifications
4. Baling

2.5 Cis-NR vs Trans-NR (Gutta-percha)

cis-1,4-Polyisoprene (Natural rubber): Soft, highly elastic at room temperature.

trans-1,4-Polyisoprene (Gutta-percha / balata): Harder, more crystalline, lower elasticity. Used historically for golf balls, insulation, dental applications.

The geometric isomerism around the double bond is responsible for the large difference in properties.

2.6 Latex Industries and Kerala Rubber Industry

Kerala accounts for the major share of India’s natural rubber production. Important industries:

- Latex products: gloves, balloons, catheters, condoms, foam mattresses, adhesives.
- Dry rubber products: tyres, tubes, footwear (Hawaii slippers), conveyor belts, hoses, moulded goods.

The Rubber Board provides technical support, quality control and market promotion. Many small and medium enterprises in Kottayam, Ernakulam and Pathanamthitta districts process latex and dry rubber.

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പ്രകൃതിദത്ത റബ്ബർ പ്രധാനമായും ഹീവിയ ബ്രസിലിയൻസിസ് മരത്തിൽ നിന്നുള്ള ലാറ്റക്സിൽ നിന്നാണ് ലഭിക്കുന്നത്. കേരളം ഇന്ത്യയിലെ പ്രധാന റബ്ബർ ഉൽപാദക സംസ്ഥാനമാണ്. റബ്ബർ ബോർഡ് കോട്ടയത്താണ്. cis-പോളിഐസോപ്രീൻ ആണ് NR; trans രൂപം ഗട്ടാ-പെർച്ച. RSS, TSR, ISNR എന്നിവ പ്രധാന ഡ്രൈ റബ്ബർ രൂപങ്ങളാണ്.

Practical Module II

Activities

- Prepare / observe samples of RSS and ISNR; note visual and tactile differences.
- Demonstrate coagulation of NR latex (acid addition → gel formation → sheet).
- Demonstrate preservation of NR latex (ammonia addition, observation of stability).
- Prepare album of 6 NR products: Tyre, Inner tube, Latex gloves, Balloon, Hawaii slippers, Latex foam / sponge. Record properties and applications for each.

Lab Series Exam – I

Practical test covering the above experiments (observation, procedure, viva).

Precautions

- Handle ammonia carefully (irritant); work in ventilated area.
- Acid coagulation: use dilute acids; wear gloves and goggles.
- Dispose of latex waste properly.

Module III — Synthetic Rubber

Instructional Hours: Theory 11 h · Practical 11 h · Mapped CO: CO3

3.1 Introduction to Synthetic Rubber

Synthetic rubbers are elastomers manufactured by polymerisation of petrochemical monomers. They were developed to overcome limitations of natural rubber (supply, oil resistance, temperature resistance, etc.).

General purpose (GP): Used in large volume for tyres and general goods – SBR, BR, IR.

Special purpose (SP): Designed for specific properties – NBR (oil resistance), CR (weather/flame), IIR (gas impermeability), EPDM (weather), CSM, Silicone (extreme temperature).

3.2 IR – Isoprene Rubber (Synthetic Polyisoprene)

Raw material: Isoprene (from petrochemical C5 stream).

Structure: Predominantly cis-1,4-polyisoprene (similar to NR but may have lower cis content depending on catalyst).

Five properties: High resilience, good tensile strength, low hysteresis, similar to NR, good building tack.

Applications: Tyres (sidewalls, carcass), medical goods, adhesives, footwear – where NR-like properties are needed but consistency is preferred.

3.3 SBR – Styrene-Butadiene Rubber

Raw materials: Styrene + Butadiene (random copolymer).

Structure: Random copolymer; styrene content typically 23–25 % for tyre grades.

Five properties: Good abrasion resistance, good aging, moderate resilience, lower cost than NR, processable.

Applications: Passenger car tyre treads, conveyor belts, footwear, hoses, adhesives.

3.4 BR – Butadiene Rubber (Polybutadiene)

Raw material: 1,3-Butadiene.

Structure: High-cis or medium-cis polybutadiene (depends on catalyst).

Five properties: Excellent resilience, low glass-transition temperature, good abrasion resistance, low heat build-up, poor wet grip (often blended).

Applications: Tyre treads and sidewalls (blended with SBR/NR), golf balls, impact modifiers for plastics.

3.5 IIR – Isobutylene-Isoprene Rubber (Butyl Rubber)

Raw materials: Isobutylene + small amount of isoprene.

Structure: Mostly saturated polyisobutylene chain with occasional unsaturation for vulcanisation.

Five properties: Outstanding gas impermeability, good weather/ozone resistance, high damping, chemical resistance, low resilience.

Applications: Inner tubes, tubeless tyre inner liners, pharmaceutical stoppers, vibration mounts, sealants.

3.6 EPDM – Ethylene-Propylene-Diene Monomer Rubber

Raw materials: Ethylene + Propylene + small diene (ENB or DCPD).

Structure: Saturated main chain (excellent weather resistance) with side-chain unsaturation for vulcanisation.

Five properties: Excellent ozone/weather/heat resistance, good electrical insulation, water resistance, moderate oil resistance, colourable.

Applications: Automotive weather-strips, roofing membranes, cable insulation, radiator hoses, window seals.

3.7 NBR – Nitrile Rubber (Acrylonitrile-Butadiene Rubber)

Raw materials: Acrylonitrile + Butadiene.

Structure: Random copolymer; ACN content 18–50 % controls oil resistance.

Five properties: Excellent oil and fuel resistance, good abrasion, moderate heat resistance, poor ozone resistance (needs protection), good tensile strength.

Applications: Oil seals, O-rings, fuel hoses, gaskets, printing rolls, gloves (industrial).

3.8 CR – Chloroprene Rubber (Neoprene)

Raw material: Chloroprene (2-chloro-1,3-butadiene).

Structure: Predominantly trans-1,4-polychloroprene.

Five properties: Good oil resistance, excellent weather/ozone/flame resistance, good mechanical properties, moderate heat resistance, self-extinguishing tendency.

Applications: Wetsuits, wire & cable jackets, gaskets, adhesives, belts, hoses, vibration mounts.

3.9 CSM – Chlorosulphonated Polyethylene

Raw material: Polyethylene modified by chlorination and chlorosulphonation.

Structure: Polyethylene backbone with Cl and $–SO_2Cl$ groups.

Five properties: Outstanding ozone/weather resistance, good chemical resistance, colour retention, flame resistance, moderate oil resistance.

Applications: Cable sheathing, coated fabrics, pond liners, automotive hoses, protective coatings.

3.10 Silicone Rubber

Raw material: Organosiloxanes (dimethylsiloxane etc.).

Structure: –Si–O– inorganic backbone with organic side groups (usually methyl).

Five properties: Extreme temperature resistance (–60 °C to +250 °C), excellent weather/ozone resistance, physiological inertness, good electrical insulation, moderate mechanical strength.

Applications: Medical implants & tubing, kitchenware, seals in extreme environments, electrical insulation, baby products, aerospace seals.

മലയാളം വിശദീകരണം

സിന്തറ്റിക് റബ്ബറുകൾ പെട്രോകെമിക്കൽ മോണോമറുകളിൽ നിന്ന് നിർമ്മിക്കുന്നു. SBR, BR, IR ജനറൽ പർപ്പസ്; NBR ഓയിൽ റെസിസ്റ്റൻസ്; CR വെതർ/ഫ്ലേം; IIR ഗ്യാസ് ഇമ്പെർമിബിലിറ്റി; EPDM വെതർ; സിലിക്കോൺ അത്യധികം താപനില റെഞ്ച്.

Comparison of Major Synthetic Rubbers

Rubber	Key Strength	Main Limitation	Typical Use
SBR	Abrasion, cost	Resilience lower than NR	Tyre treads
BR	Resilience, low T_g	Wet grip	Tyre blends
IIR	Gas barrier	Low resilience	Inner liners
EPDM	Weather/ozone	Oil resistance	Weather-strips
NBR	Oil/fuel	Ozone	Seals, hoses
CR	Weather + oil + flame	Cost	Wetsuits, cables
Silicone	Temperature extremes	Mechanical strength, cost	Medical, high-temp seals

Practical Module III

Collect samples of products made from:

- SBR, BR
- IIR, EPDM
- NBR, CR
- CSM, Silicone rubber

For each sample: label the polymer name, list three properties and three applications. Prepare a neat album. Observe differences in feel, flexibility, colour and typical end-use.

Module IV — Plastics and Fibres

Instructional Hours: Theory 11 h · Practical 11 h · Mapped CO: CO4

4.1 Classification of Plastics — Thermoplastics and Thermosets

Thermoplastics: Soften on heating and harden on cooling; process can be repeated. Linear or branched chains held by secondary forces.

Five examples: Polyethylene (PE), Polypropylene (PP), Polyvinyl chloride (PVC), Polystyrene (PS), Nylon (polyamide), PET, ABS.

Thermosets: Undergo irreversible chemical cross-linking on heating (or curing); cannot be remelted. Three-dimensional network.

Five examples: Bakelite (phenol-formaldehyde), Melamine-formaldehyde, Urea-formaldehyde, Epoxy resins, Unsaturated polyesters, Vulcanised rubber (elastomeric thermoset).

Five properties comparison

Property	Thermoplastics	Thermosets
Melting / softening	Reversible	Irreversible (degrade)
Solubility	Often soluble in solvents	Insoluble
Recyclability	Generally recyclable	Difficult to recycle
Dimensional stability at high T	Lower	Higher
Impact resistance	Often higher (tough)	Often more brittle

Advantages of Plastics

- Lightweight, corrosion resistant, easily moulded into complex shapes, electrical insulation, wide range of properties, low cost for many grades.

Disadvantages

- Non-biodegradability (most), limited high-temperature performance (many thermoplastics), environmental concerns (microplastics, waste), some release toxic fumes on burning.

4.2 Common Household Articles and the Plastics Used

Kitchen and Food Storage

- Water / milk bottles → PET, HDPE, PC (older)
- Food containers (Tupperware-type) → PP, HDPE, LDPE
- Sandwich bags, plastic wrap → LDPE, PVC cling film
- Cutlery, microwave-safe dishes → PP, PS
- Bottle caps → PP, HDPE
- Dustbins, buckets, straws → HDPE, PP, PS

Bathroom and Personal Care

- Toothbrushes → PP handle, Nylon bristles
- Cosmetic containers → PET, PP, ABS
- Razors → PS, ABS, PP
- Shower curtains → PVC, PE
- Shampoo / soap bottles → HDPE, PET, PP

Cleaning and Storage

- Spray bottles → PET, HDPE, PP
- Trash bags → LDPE, HDPE
- Storage bins → PP, HDPE
- Hoses, pipes → PVC, PE

Electronic and General Goods

- Mobile phone covers → TPU, PC, ABS, Silicone
- Toy components → ABS, PE, PP
- Pens, notebook covers → PP, ABS, PS
- Packaging for online orders → PE films, EPS, PP
- Internal electronic components → PC, ABS, PBT, epoxy

Furniture

- Plastic chairs, tables → PP, HDPE, ABS
- Storage crates → HDPE, PP
- Insulation sheets → EPS, XPS, PU foam

4.3 Fibres

Definition: Long, thin, flexible structures with high length-to-diameter ratio, used for textiles, ropes, composites, etc.

Classification

- **Natural fibres**: From plants or animals – cotton, wool, silk, coir, jute, arecanut fibre, flax.
- **Synthetic fibres**: Man-made from polymers – Nylon, Polyester, Acrylic, Polypropylene, Polyethylene (UHMWPE), Carbon fibre, Glass fibre, Aramid.

Properties and Applications of Selected Fibres

Fibre	Key Properties	Applications
Wool	Warm, resilient, moisture absorbing	Clothing, carpets, blankets
Silk	Lustrous, strong, smooth	Luxury textiles, medical sutures
Coir	Coarse, durable, salt-water resistant	Mats, ropes, geo-textiles, mattresses
Arecanut	Natural, biodegradable	Composite boards, eco-products
Nylon	High strength, abrasion resistant, elastic	Apparel, ropes, tyres, carpets
Acrylic	Wool-like, colourfast, lightweight	Sweaters, blankets, awnings
PE / PP	Lightweight, chemical resistant, low cost	Ropes, non-wovens, geotextiles
Carbon fibre	Extremely high strength & stiffness, light	Aerospace, sports, automotive composites

Fibre	Key Properties	Applications
Glass fibre	High strength, insulating, low cost	Composites, insulation, PCBs

Many synthetic fibres are made from the same polymers used as plastics (Nylon, PET, PP) but processed by spinning into continuous filaments or staple fibres.

4.4 Liquid Resins, Adhesives and Sealants

- **Liquid resins**: Reactive liquid polymers that cure to solid. Examples: Epoxy resin, unsaturated polyester resin, polyurethane resin.
- **Adhesives**: Materials that join surfaces. Examples: PVA (white glue), epoxy adhesive, cyanoacrylate (super glue), rubber-based adhesives.
- **Sealants**: Fill gaps and prevent passage of fluids/gases. Examples: Silicone sealant, polyurethane sealant, acrylic sealant.

മലയാളം വിശദീകരണം

തെർമോപ്ലാസ്റ്റിക്ക് ചൂടാക്കുമ്പോൾ മൃദുവായി; വീണ്ടും ഉപയോഗിക്കാം. തെർമോസെറ്റുകൾ ക്യൂർ ചെയ്താൽ വീണ്ടും ഉരുകില്ല. ഫൈബറുകൾ പ്രകൃതിദത്തവും കൃത്രിമവുമാണ്. നൈലോൺ, പോളിസ്റ്റർ, കാർബൺ ഫൈബർ എന്നിവ പ്രധാന സിന്തറ്റിക് ഫൈബറുകളാണ്.

Practical Module IV

- Collect 3 samples of raw thermoplastics and 3 of thermosets; label polymer name.
- Collect 3 product samples each of LDPE, HDPE and PVC; record name, 3 properties and application.
- Collect 3 natural fibre samples and 3 synthetic fibre samples; label name, 3 properties and application.
- Prepare a systematic album.

Lab Series Exam – II covers the above practical work.

Suggested Student Activities for Self-Learning

Total Self-Learning hours: 90 (6 h per week × 15 weeks). Each activity reinforces the corresponding module.

Module I (Weeks 1–4)

1. **Week 1**: Explain the relationship between molecular structure (monomers, repeating units) and material properties such as strength, flexibility and transparency. Collect 2 dried samples of natural polymer and learn their 3 main properties and applications from reliable sources.
2. **Week 2**: Illustrate the conceptual representation of polymer molecules by linking simple molecules. Collect 2 dried samples of synthetic polymer and study 3 main properties and applications.
3. **Week 3**: Explore what Bakelite is (first fully synthetic plastic). Collect 5 samples of organic polymers and learn 3 main properties and applications each.
4. **Week 4**: Find 10 monomers and list the polymers made from them. Collect a sample of inorganic polymer (silicone) and learn its 3 main properties and applications.

Module II (Weeks 5–8)

5. **Week 5**: Explore different trees from which latex can be obtained. Collect samples of RSS and ISNR and learn the difference in properties.
6. **Week 6**: Compare creaming and centrifuging methods. Search different preservatives used for NR latex.
7. **Week 7**: Find centrifuging units in Kerala. Visit (or virtually study) a latex industry and a dry rubber industry. Learn the chemical reaction during coagulation of NR latex.
8. **Week 8**: Learn the chemical name and structure of Natural rubber. Understand the difference between a latex product and a dry rubber product.

Module III (Weeks 9–12)

9. **Week 9**: Search different products made of synthetic rubbers. Collect SBR and BR product samples; study 3 properties and 3 applications each.
10. **Week 10**: Learn structure of CSM and Silicone rubber. Collect IR and EPDM product samples; study properties and applications.
11. **Week 11**: Learn structure of SBR, IR and BR. Collect NBR and CR product samples; study properties and applications.
12. **Week 12**: Learn structure of EPDM, NBR and CR. Collect CSM and Silicone product samples; study properties and applications.

Module IV (Weeks 13–15)

13. **Week 13**: Find 10 different products made of thermoplastics and thermosets. Collect 3 raw thermoplastic and 3 thermoset samples; study name, properties and applications.
14. **Week 14**: Learn properties required for fibre-forming polymers. Collect 3 natural fibre samples; study name, properties and applications.
15. **Week 15**: Learn applications of fibres. Collect 3 synthetic fibre samples; study name, properties and applications.

Learning Resources

Text Books

Sl. No.	Title	Author	Publication
1	Polymer Science	Jayadev Sreedhar, V. R. Gowariker, N. V. Viswanathan	New Age International (P) Ltd, New Delhi – 2005 Edition
2	Text Book of Polymer Science	Billmeyer F. W. (J.R.)	John Wiley & Sons (Asia) PTE Ltd, Singapore, 2008 Edition
3	Man Made Fibres	R. W. Moncrieff	Wiley, 5th edition, 1970
4	Handbook of Natural Rubber Production – Rubber Board	RRI, Kottayam	—

References

Sl. No.	Title	Author	Publication
1	Introductory Polymer Chemistry	G. S. Misra	New Age International (P) Ltd, New Delhi – 2005 Edition
2	Introduction to Polymer Science and Technology	Mustafa Akay	bookboon.com

Web Resources

Sl. No.	URL	Modules Covered
1	https://study.com	I
2	https://www.osborneindustries.com	III
3	https://www.sealsdirect.co.uk	II
4	https://www.sciencedirect.com	All

Major Equipment / Software

As per the official syllabus: **NIL** special major equipment or software is listed for a batch of 30 students. Practical work is based on sample collection, demonstration of coagulation/preservation and album preparation.

Assessment Methodology – Practicum

Overall Scheme

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Avg of Self-learning	Continuous Evaluation	Practical test (1st half)	Practical test (2nd half)	Series Exam (2 modules)	Series Exam (2 modules)	Written (Theory)	Lab Exam (Internal)
Duration	—	—	—	3 h	3 h	2 h	2 h	2.5 h	3 h
Exam marks	—	10	50	40	40	50	50	60	40
Converted marks	5	5	10	10	10	10	10	60	40
Total ESE + CIE related = 100 (Theory 60 + Lab 40) + Internal components									

Tentative schedule: CA1 by 4th week, CE by 7th week, CA2 by 11th, CA3 by 14th, Series 1 in 8th week, Series 2 in 15th week, ESE at end of semester.

Scheme of Evaluation – Practical (Continuous Internal Evaluation) – 50 marks

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely record submission, pre-lab preparation, understanding of objectives	10
2	Experimental Setup & Procedure	Identification of materials, systematic and safe execution	10
3	Observation & Data Recording	Accuracy, neatness, units, tabulation	5
4	Analysis & Result Interpretation	Proper discussion, logical conclusion	10
5	Viva Voce / Concept Understanding	Clarity of concepts, ability to answer related questions	10
6	Workmanship, Discipline & Teamwork	Lab discipline, care of materials, teamwork, safety	5
	Total		**50**

Scheme of Evaluation – Practical Test – 40 marks

Category	Things to Evaluate	Marks
Write-up / Procedure	Aim, procedure steps	10
Experiment Setup & Execution	Correct handling, accuracy	10
Observation & Result	Tabulation, correctness of outcome	8
Viva Voce	Conceptual understanding	8
Record	Completion & neatness, certification	4
Total		**40**

Series Examination Pattern – Theory (2 hours, 50 marks)

Any two modules per series. Part A: 4 questions × 1 mark. Part B: 6 questions × 3 marks. Part C: 4 questions × 7 marks (with choice). Total 50.

End Semester Examination – Theory (2.5 hours, 60 marks)

Module	Part A (2 Q from each, 1 mark)	Part B (2 out of 3 from each, 3 marks)	Part C (1 set with choice, 7 marks)	Total
1	2	$2 \times 3 = 6$	7	15
2	2	$2 \times 3 = 6$	7	15
3	2	$2 \times 3 = 6$	7	15
4	2	$2 \times 3 = 6$	7	15
Total	8	24	28	**60**

Question Bank (with Answers)

Module I – Sample Questions

▼ 1. Define polymer and monomer. Give two examples of each. (2 marks)

Answer: A monomer is a small molecule that can join with others to form a polymer. Examples: ethene, isoprene. A polymer is a large molecule made of many repeating monomer units. Examples: polyethylene, natural rubber (polyisoprene).

▼ 2. State the relationship between Degree of Polymerisation and molecular weight. (2 marks)

Answer: M_w (polymer) = DP × M_0 (repeating unit), where DP is the average number of repeating units.

▼ 3. Distinguish between thermoplastics and thermosets with two examples each. (3 marks)

Answer: Thermoplastics soften on heating and can be reprocessed (PE, PVC). Thermosets form irreversible cross-links and cannot be remelted (Bakelite, epoxy).

▼ 4. A polymer has average MW 56 000 and repeating unit MW 28. Calculate DP. (3 marks)

Answer: DP = 56 000 / 28 = 2000.

▼ 5. Explain classification of polymers based on structure (linear, branched, cross-linked). (7 marks)

Answer: Linear – continuous chain, pack well, higher crystallinity (HDPE). Branched – side chains, lower density (LDPE). Cross-linked – 3-D network, insoluble, non-melting (vulcanised rubber, Bakelite). Include sketches and property consequences.

Module II – Sample Questions

▼ 1. What is the monomer of natural rubber? Write its structure. (2 marks)

Answer: Isoprene (2-methyl-1,3-butadiene), $CH_2=C(CH_3)–CH=CH_2$.

▼ 2. Differentiate field latex and concentrated latex. (3 marks)

Answer: Field latex – fresh from tree, DRC ~30–35 %, unstable. Concentrated latex – DRC ~60 % by centrifuging/creaming, more stable and transportable.

▼ 3. Outline the production of RSS. (7 marks)

Answer: Dilution → sieving → acid coagulation → milling into sheets → ribbing → smoking in smoke house → grading (RSS 1–5) → packing. Explain each step briefly.

▼ 4. Why is cis configuration important in natural rubber? (3 marks)

Answer: Cis-1,4 configuration gives a flexible chain that can coil and uncoil easily, producing high elasticity. Trans configuration (gutta-percha) is more crystalline and less elastic.

Module III – Sample Questions

▼ 1. Name two general-purpose and two special-purpose synthetic rubbers. (2 marks)

Answer: GP – SBR, BR (or IR). SP – NBR, CR (or IIR, EPDM, Silicone).

▼ 2. Why is NBR preferred for oil seals? (3 marks)

Answer: NBR has polar acrylonitrile groups that resist swelling in non-polar oils and fuels; oil resistance increases with ACN content.

▼ 3. Compare SBR and Natural Rubber for tyre applications. (7 marks)

Answer: SBR – better abrasion and aging, lower cost, poorer resilience and building tack. NR – higher resilience, better tear strength and tack. Blends are commonly used.

▼ 4. List five properties and three applications of Silicone rubber. (5 marks)

Answer: Properties – extreme temperature range, weather/ozone resistance, biocompatibility, electrical insulation, flexibility. Applications – medical tubing, high-temp seals, kitchenware.

Module IV – Sample Questions

▼ 1. Give five examples each of thermoplastics and thermosets. (3 marks)

Answer: Thermoplastics – PE, PP, PVC, PS, Nylon. Thermosets – Bakelite, melamine-formaldehyde, epoxy, unsaturated polyester, urea-formaldehyde.

▼ 2. Which plastic is commonly used for water bottles and why? (2 marks)

Answer: PET – clear, tough, good barrier to gases, recyclable, food-safe.

▼ 3. Distinguish natural and synthetic fibres with examples and one property each. (5 marks)

Answer: Natural – cotton (absorbent), wool (warm), silk (lustrous), coir (durable). Synthetic – Nylon (strong, elastic), Polyester (crease-resistant), Acrylic (wool-like), Carbon (high stiffness).

▼ 4. What are liquid resins, adhesives and sealants? Give two examples of each. (5 marks)

Answer: Liquid resins – epoxy, unsaturated polyester. Adhesives – PVA, cyanoacrylate. Sealants – silicone sealant, polyurethane sealant.

Viva / Lab Questions (Sample)

- How do you preserve NR latex in the laboratory?
- What acid is commonly used for coagulation of latex?
- How can you distinguish LDPE from HDPE by simple observation?
- Why is silicone rubber preferred for medical applications?
- Name the polymer used in most plastic chairs.
- What is the repeating unit of polyethylene?
- Why are thermosets not recycled by simple melting?

Quick Revision

Module I – One-Page Summary

- Polymers = macromolecules from monomers via polymerisation.
- DP = M_w / M_0.
- Classification: origin (natural/synthetic/derived), backbone (organic/inorganic, homo/hetero chain), monomer (homo/co – random/alternating), architecture (linear/branched/cross-linked).
- Hydrocarbons: saturated (alkanes), unsaturated (alkenes, alkynes), aliphatic vs aromatic.

Module II – One-Page Summary

- NR = cis-1,4-polyisoprene from Hevea latex.
- Latex forms: field → concentrated (centrifuge/cream) → double centrifuged.
- Preservation: ammonia.
- Dry forms: RSS (smoked sheets), TSR/ISNR (technically specified).
- Cis = elastic; Trans (gutta-percha) = harder.
- Kerala + Rubber Board = centre of Indian NR industry.

Module III – One-Page Summary

- GP: SBR, BR, IR – tyres & general goods.
- SP: NBR (oil), CR (weather+oil+flame), IIR (gas barrier), EPDM (weather), CSM, Silicone (temp extremes).
- Always remember the key property that decides the application.

Module IV – One-Page Summary

- Thermoplastics = remeltable; Thermosets = permanent network.
- Household plastics: PET bottles, PP containers, HDPE pipes/buckets, PVC pipes/curtains, PS cutlery, ABS toys.
- Fibres: natural (wool, silk, coir...) vs synthetic (Nylon, Acrylic, Carbon, Glass...).
- Liquid resins, adhesives, sealants – curing systems for joining and sealing.

Formula Sheet

$$M_w = DP \times M_0$$

$$DP = M_w / M_0$$

Common Mistakes & Exam Tips

- Do not confuse cis-NR with trans (gutta-percha).
- Thermoplastics can be reshaped; thermosets cannot – remember with examples.
- For synthetic rubbers, always link the special property to the application (NBR→oil, IIR→air retention, etc.).
- In numericals, check units of molecular weight (g/mol).
- Album work: neat labels, three properties + applications for every sample.
- Read the question carefully – “list five properties” vs “explain with structure”.

Glossary of Important Terms

- **Monomer** – Small molecule that can polymerise.
- **Polymer** – Large molecule of repeating units.
- **Degree of Polymerisation (DP)** – Average number of repeating units.
- **Homopolymer** – Polymer from one monomer type.
- **Copolymer** – Polymer from two or more monomer types.
- **Vulcanisation** – Cross-linking of rubber (usually with sulphur).
- **Latex** – Aqueous dispersion of polymer particles (NR latex).
- **Coagulation** – Conversion of latex into solid rubber by acid or other agents.
- **Thermoplastic** – Softens reversibly on heating.
- **Thermoset** – Forms permanent network on curing.
- **Elastomer** – Polymer that can stretch largely and recover (rubber).
- **Fibre** – Long thin structure with high aspect ratio used in textiles/composites.
- **Repeating unit** – Structural unit that repeats in the polymer chain.
- **Cross-link** – Covalent bridge between polymer chains.
- **Glass transition temperature (T_g)** – Temperature below which polymer becomes glassy/brittle.